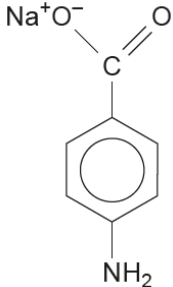
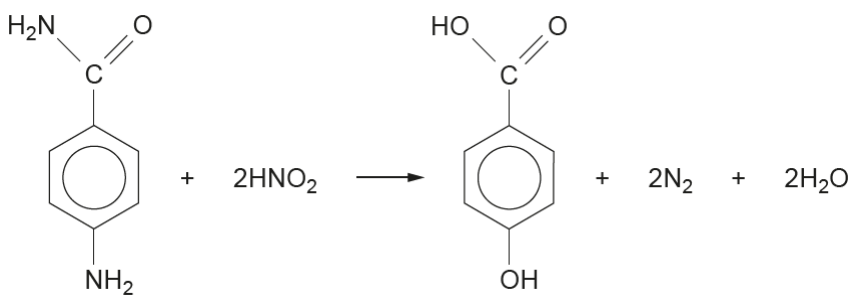


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			Indicative content molar mass is 72 g mol⁻¹ of which 50.0% is carbon ⇒ each molecule of compound G must contain 3 carbon atoms this leaves a mass of 36 ⇒ must be 2 oxygen atoms and 4 hydrogen atoms as 1 oxygen atom and 20 hydrogen atoms is not feasible molecular formula is C₃H₄O₂ it reacts with Tollens reagent ⇒ must be an aldehyde / have the CHO group it gives a yellow solid with alkaline iodine ⇒ must have a $\text{CH}_3-\text{C}(=\text{O})$ group or a $\text{CH}_3-\text{C}(\text{OH})(\text{H})$ group only 2 hydrogen environments, CH₃ group (at ~2.2) and CHO group (at ~9.4) ⇒ suggests $\text{CH}_3-\text{C}(=\text{O})-\text{C}(=\text{O})\text{H}$ reduction of this compound gives a diol with molar mass 76 g mol⁻¹ $\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{OH} \\ \quad \\ \text{OH} \quad \text{H} \end{array}$ ⇒ confirms above structure	2	2	2	6		2

Question				Marking details
				5-6 marks All the information has been used including the NMR spectrum; correct structure given <i>The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary is used accurately throughout.</i> 3-4 marks Most of the information has been used correctly but there are some omissions; some correct features in the structure <i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i> 1-2 marks Some of the information has been used but there are many omissions <i>The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			$E_1 \times \lambda_1 = E_2 \times \lambda_2$ (1) $\lambda_2 = \frac{E_1 \times \lambda_1}{E_2} = 267 \text{ kJ mol}^{-1}$ (1) alternative method constant = $E \times \lambda = 1.2 \times 10^5$ (1) $E = \frac{1.2 \times 10^5}{450} = 267 \text{ kJ mol}^{-1}$ (1)		1	1	2	1	
	(c)	(i)	I	 accept if Na ⁺ not included			1	1		
			II	the attacking reagent / OH ⁻ is a nucleophile(1) lone pair on N becomes part of delocalised system / C—N bond is stronger when directly attached to ring (1)		2		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		 correct formulae (1) balancing (1)				2		
				Question 10 total	2	7	4	13	1	2